Magnet Mapping

John Haggerty

In the beginning…

ASG SUPERCONDUCTORS magnets & systems unit

MAGNETS FOR FUSION

MAGNETS FOR HIGH ENERGY PHYSICS

MAGNETS FOR MEDICAL APPLICATIONS

SYSTEMS FOR ENERGY

SERVICES & REPAIRS

INFN/SLAC/LLNL
BABAR

BaBaR Experiment Thin Solenoid

The 1.5 T thin superconducting solenoid (peak field 2.3 T in the conductor), completed in 1998, is part of the BaBar detector, located on the PEP-II B-Factory machine at the Stanford Linear Accelerator Center (USA).
The solenoid measures 3.38 m at the inner bore diameter, 3.7 m in length. The coil, 1067 turns of Al stabilised NbTi conductor, is wound on two layers inside an Al mandrel. The winding is axially graded to produce the requested ±3% field uniformity in the tracking region; the stored energy is 27 MJ, the inductance 2.57 H and the operating current 4596 A.
The coil has been isolated with multiple layers of glass cloth, impregnated in epoxy resin using the VPI process. The coil is indirectly cooled at an operating temperature of 4.5 K using the thermo-siphon technique. The liquid helium is circulated in channels welded to the Al mandrel.

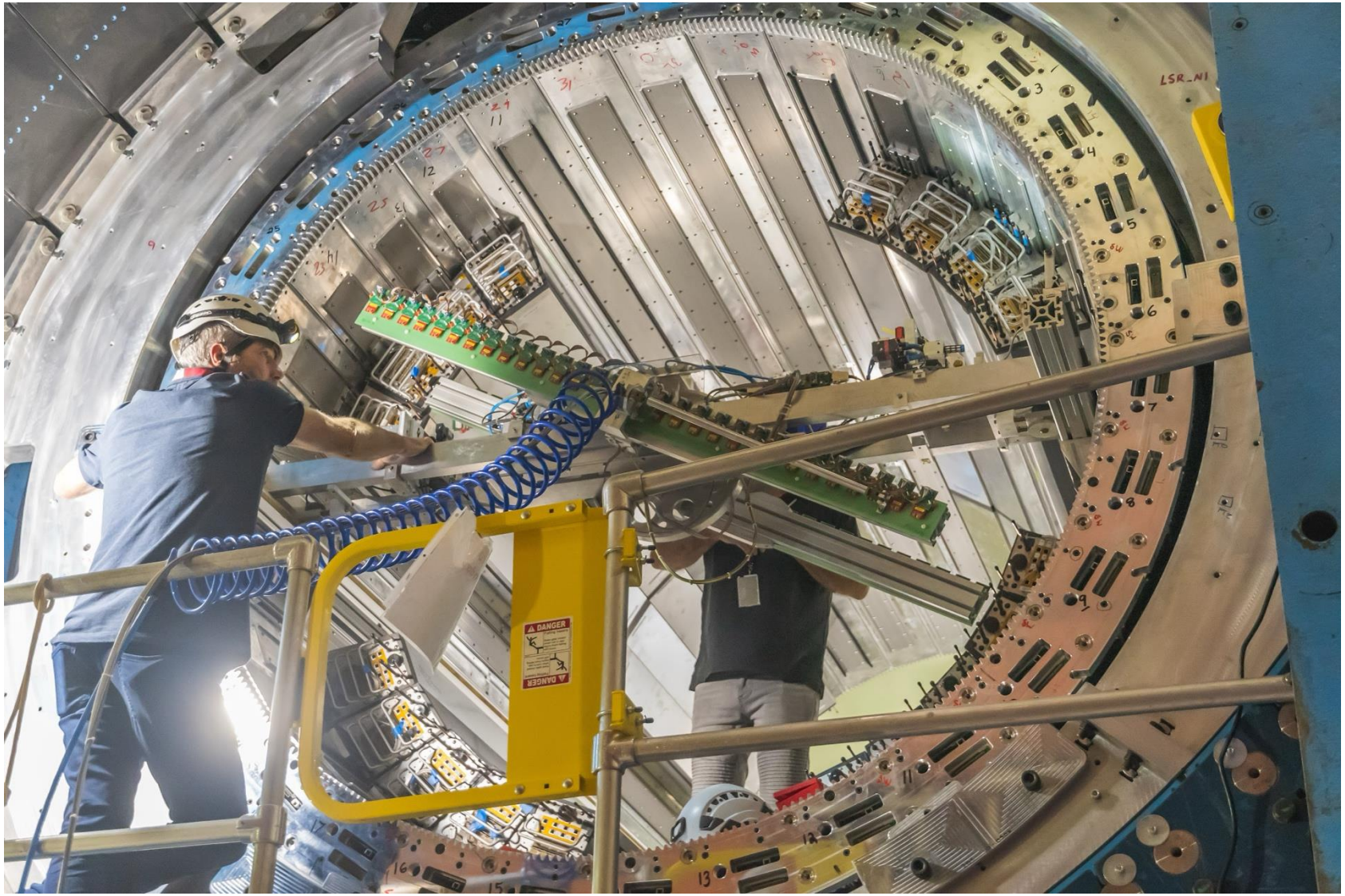
Main parameters

Maximum field	1.5 T
Stored energy	27 MJ
Inner Diameter	ϕ 3380 mm
Magnet Length	3700 mm
Inductance	2.5 H
Nominal current	4596 A
Conductor	NbTi – Al
coextruded	
Type of Winding	outer mandrel
Type of cooling	indirect

www.asgsuperconductors.com

ASG Superconductors s.p.a. all rights reserved 2019

https://www.asgsuperconductors.com/progetto/babar

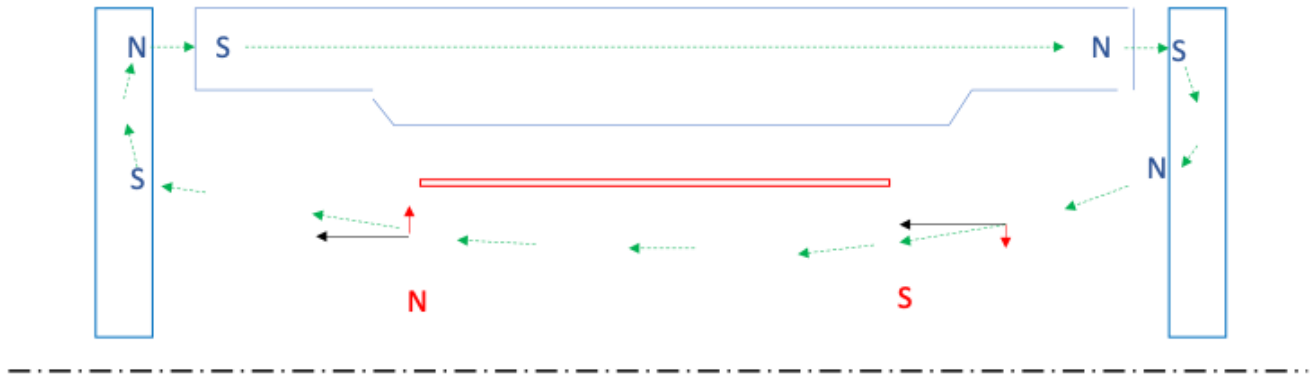


CERN group did the mapping October 31-November 11, 2022

Solenoidal sanity checks

- The field in the “center” is pretty uniform and B_z is maximal
- B_ρ changes direction at the “center” (see figure)
- B_ϕ is zero
- $\vec{B}$ can be tilted from the beam axis (but not too much)

Black Arrow = Bz Red Arrow = Br Green Arrow = B_total (flux)



From Wuzheng Meng

Just the facts

```
/gpfs/mnt/gpfs02/sphenix/data/data02/sphenix/MagnetMapping/cernfinal
[jsh@sphnxuser08 cernfinal]$ ls -l
total 99585
-rw-r--r--+ 1 jsh sphenix 50667520 Nov 11  2022 download.tar
-rw-r--r--+ 1 jsh sphenix 35765363 Nov 10  2022 fieldMapFineFullField.csv
-rw-r--r--+ 1 jsh sphenix  7448956 Nov 10  2022 fieldMapRoughFullField.csv
-rw-r--r--+ 1 jsh sphenix  7448793 Nov 10  2022 fieldMapRoughHalfField.csv
-rw-r--r--+ 1 jsh sphenix      371 Nov 10  2022 Readme.txt
```

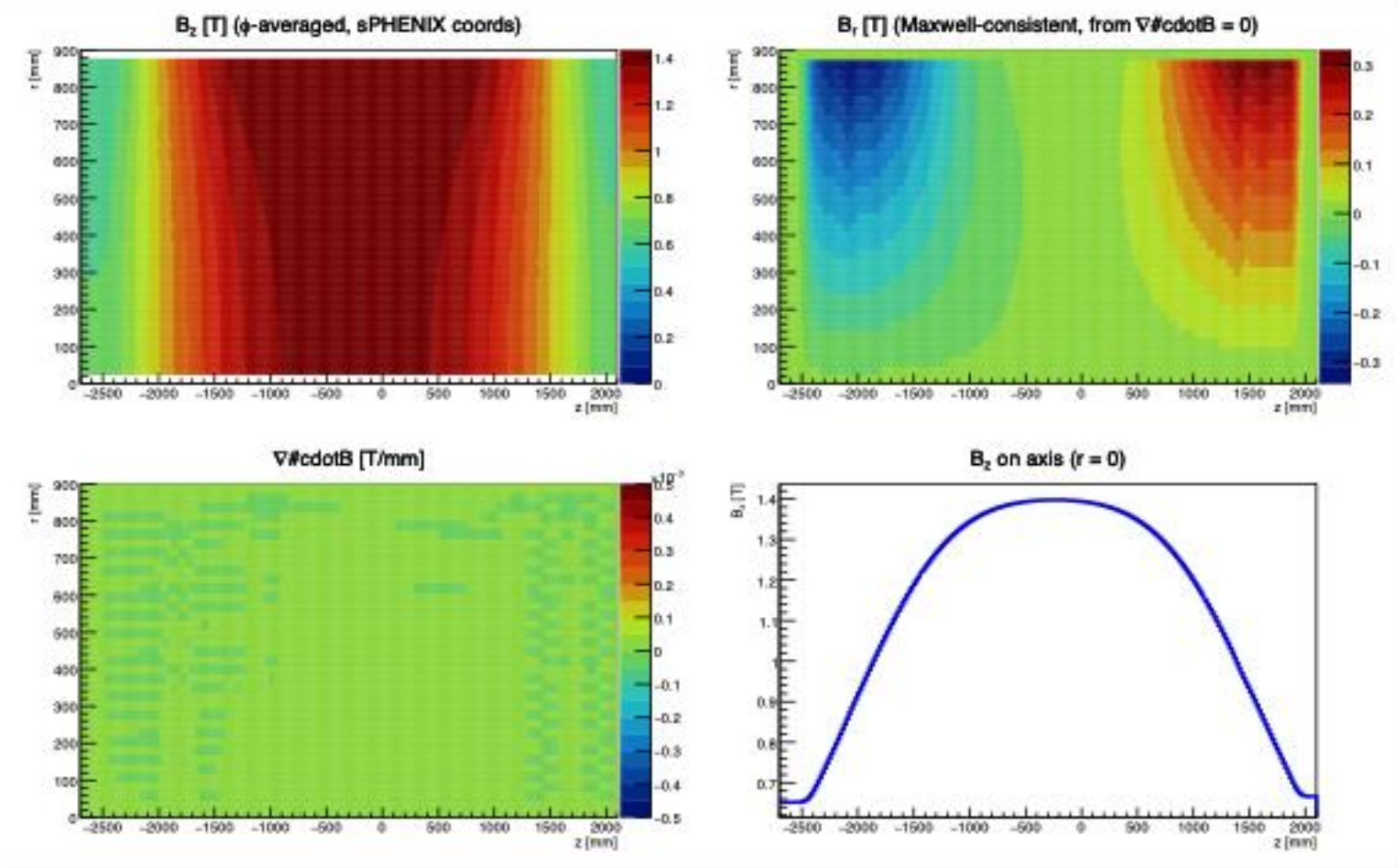
- x,y,z are in mm
- b, bx, by, bz are in Tesla
- x,y,z are in "survey coordinates:"
  - y is south (our -z)
  - z is up (our y)
  - x is west (to be right handed, same as our x)
- The structure in the steps in y is due to the fact that there are two measurement planes

Analysis

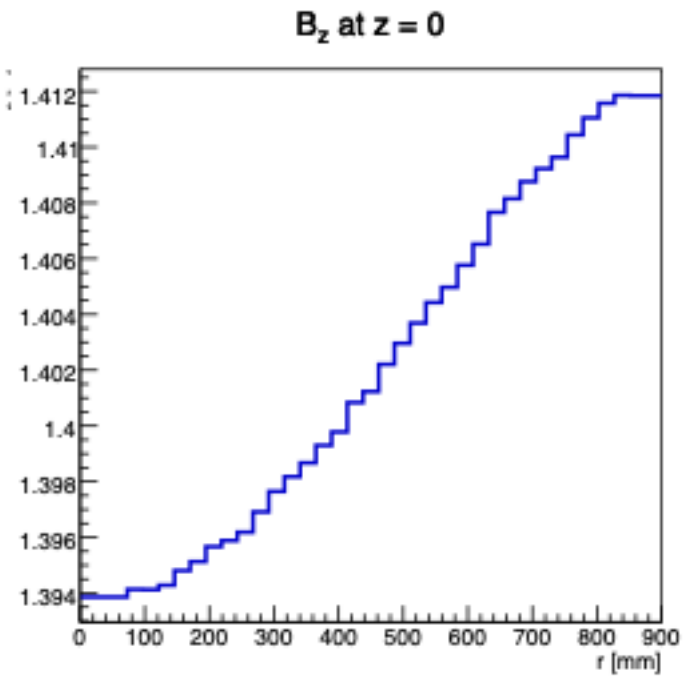
- I can't 100% vouch for what follows, but it looks about right to me
- I worked with my new best silicon friend, and we converted the coordinates to the sPHENIX coordinate system (to the best of my knowledge), and made a class that interpolated the field to any location, and constrained it so that the field obeys Maxwell's equations.
  - Claude tried a few different interpolation techniques to find the smoothest
- https://github.com/haggerty/sphenix-cernfinal-map

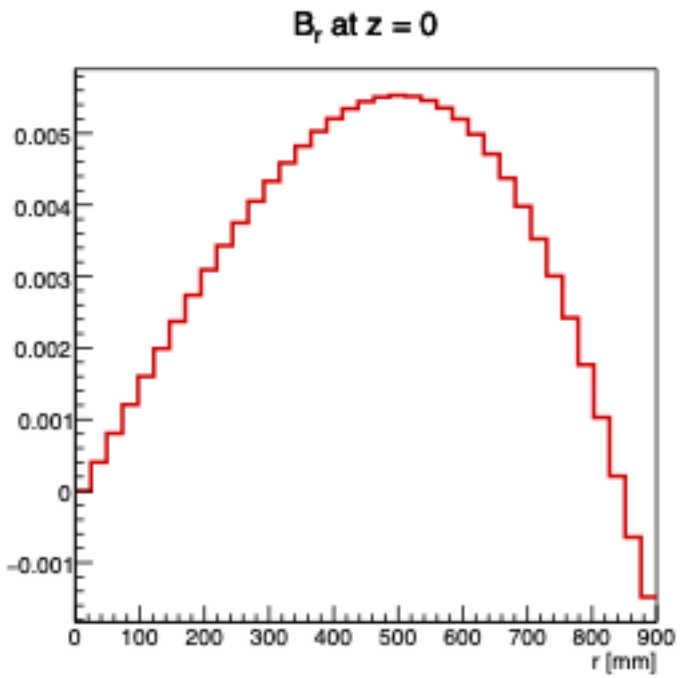
```
#include "sPHENIXFieldMap.h"

sPHENIXFieldMap fmap("fieldMapFineFullField.csv",
                     "fieldMapRoughFullField.csv");

// Cylindrical sPHENIX coordinates
double Br, Bphi, Bz;
fmap.GetField(r_mm, phi_rad, z_mm, Br, Bphi, Bz);

// Cartesian sPHENIX coordinates
double Bx, By, Bz;
fmap.GetFieldXYZ(x_mm, y_mm, z_mm, Bx, By, Bz);
```

Overview


B_z [T] (ϕ-averaged, sPHENIX coords)
B_r [T] (Maxwell-consistent, from ∇#cdotB = 0)
∇#cdotB [T/mm]
B_z on axis (r = 0)
r [mm]
z [mm]
B_z [T]

Field at midplane





Features of the map

- The central field is 1.394 T
- The magnetic “center” is at z = -240 mm
- The field is tilted from the z axis by 2.39 mr at ϕ = 71°

Next steps

- Careful comparison with Opera map now in use
- Check against survey data of the solenoid and TPC
  - Marianna is putting together a compendium of survey data (before and after)
- Further checking of internal consistency